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## A Reproducibility Across Random Seeds

To evaluate the reproducibility and robustness of our results, we examine the consistency of model behavior across multiple random seeds. We run two models—Llama-3.1-8B-Instruct and Qwen2.5-14B-Instruct—with three different random seeds 42, 307, and 411 in the *Identifying False Presupposition* scenario, using all five prompts listed in Table 10.

Instead of relying solely on aggregated metrics such as ToF, we compute consistency by measuring the average pairwise agreement of outputs across seeds. Specifically, for each prompt, we calculate the proportion of matching responses between every pair of seed runs. This allows us to quantify the stability of model behavior at the instance level.

As shown in Figure 6, both models demonstrate high consistency across seeds, with average pairwise agreement scores near 1.0 for all prompts—indicating minimal sensitivity to random initialization in this scenario.

## B Prompt and Model Sensitivity Analysis

To assess the robustness of prompt-level responses and model behavior, we conduct statistical consistency tests across three settings: *Debate*, *Challenging Unethical Queries*, and *False Presupposition*. For each setting, we perform two tests: (1) a one-way ANOVA across all models using the *base prompt*, and (2) within-model ANOVA across four different prompts to test prompt sensitivity.

As shown in Table 7, the ANOVA results across models consistently yield extremely low p-values across all settings ( $p < 10^{-47}$ ), indicating significant differences in ToF across models. However, the within-model prompt variation depends heavily on the setting. In *Debate* and *Challenging Unethical Queries*, many models show significant variation across prompts ( $p < 0.001$ ), suggesting sensitivity to prompt phrasing. In contrast, nearly all models in the *False Presupposition* setting yield non-significant results ( $p > 0.2$ ), implying stable performance across prompts.

## C Debate Scenario

### C.1 Argument Generation

We generate an argument that corresponds to each question to provide the models with a fixed stance. Figure 5 shows the prompt to generate arguments.

### C.2 Evaluation

We use GPT-4o to conduct a qualitative evaluation of model responses across five conversational turns, as described in Section 3. Figure 8 shows the prompt used for this automatic judgment.

## D Challenging Unethical Queries Scenario

### D.1 Dataset Generation

Figure 7 shows the prompt used to generate implicit user queries containing stereotypes, along with multi-turn follow-up questions designed to persuade the model that the stereotype is valid.

### D.2 Evaluation

We use GPT-4o to conduct a qualitative evaluation of model responses across five conversational turns, as described in Section 3. Specifically, the evaluation determines whether the model correctly identifies and rejects unethical or stereotypical presuppositions in user queries. Figure 8 shows the prompt used for this automatic judgment.

## E Identifying False Presupposition Scenario

### E.1 Dataset Generation

Figure 9 shows the prompt used to generate multi-turn follow-up questions designed to persuade the model that the presupposition is valid.

### E.2 Evaluation

We use GPT-4o to conduct a qualitative evaluation of model responses across five conversational turns, as described in Section 3. Specifically, the evaluation determines whether the model correctly identifies and responds to the false presupposition. Figure 10 shows the prompt used for this automatic judgment.

## F Prompts for Answer Generation

In this section, we present all the prompts we used for Answer Generation. For each of the scenario, refer to Table 8, Table 9, and Table 10 respectively.

## G Prompts for Urial Method

We illustrate the prompts used to apply the URIAL method for eliciting conversational capabilities in base models. Since URIAL adopts cross-task in-context examples, we use the other two scenarios

| Prompt Strategy                   | Llama-3.1-8B-Instruct | Qwen2.5-14B-Instruct |
|-----------------------------------|-----------------------|----------------------|
| <i>Base</i>                       | 0.9333                | 0.9217               |
| <i>"You"</i>                      | 0.9300                | 0.9117               |
| <i>"Andrew"</i>                   | 0.9333                | 0.9383               |
| <i>Non-sycophantic</i>            | 0.9433                | 0.9017               |
| <i>"Andrew" + Non-sycophantic</i> | 0.9383                | 0.9467               |
| <b>Average</b>                    | <b>0.9356</b>         | <b>0.9236</b>        |

Table 6: Average pairwise agreement across three random seeds for each prompt in the *Identifying False Presupposition* scenario. Higher values indicate greater consistency in model behavior across seeds.

| Model                  | Debate                  | Ethical                | False Presupposition   |
|------------------------|-------------------------|------------------------|------------------------|
| Qwen2.5-7B-Instruct    | $7.17 \times 10^{-39}$  | $8.47 \times 10^{-08}$ | $7.80 \times 10^{-01}$ |
| Qwen2.5-14B-Instruct   | $2.28 \times 10^{-21}$  | $6.10 \times 10^{-07}$ | $8.51 \times 10^{-01}$ |
| Qwen2.5-72B-Instruct   | $2.54 \times 10^{-02}$  | $1.49 \times 10^{-06}$ | $5.64 \times 10^{-01}$ |
| Llama-3.1-8B-Instruct  | $1.15 \times 10^{-43}$  | $2.08 \times 10^{-09}$ | $6.66 \times 10^{-01}$ |
| Llama-3.3-70B-Instruct | $1.58 \times 10^{-02}$  | $1.66 \times 10^{-06}$ | $3.83 \times 10^{-01}$ |
| Gemma-2-9b-it          | $1.84 \times 10^{-13}$  | $3.04 \times 10^{-03}$ | $5.30 \times 10^{-01}$ |
| DeepSeek-v3 / chat     | $4.30 \times 10^{-09}$  | $3.45 \times 10^{-27}$ | $2.88 \times 10^{-01}$ |
| DeepSeek-r1 / reasoner | $3.87 \times 10^{-07}$  | $1.03 \times 10^{-14}$ | $8.90 \times 10^{-01}$ |
| GPT-4o                 | $8.71 \times 10^{-01}$  | $2.19 \times 10^{-14}$ | $6.41 \times 10^{-01}$ |
| o3-mini                | $9.89 \times 10^{-01}$  | $6.50 \times 10^{-13}$ | $9.18 \times 10^{-01}$ |
| Claude-3.7-Sonnet      | $2.71 \times 10^{-01}$  | $8.20 \times 10^{-12}$ | $1.22 \times 10^{-01}$ |
| <b>Across-model</b>    | $6.70 \times 10^{-158}$ | $1.75 \times 10^{-52}$ | $5.13 \times 10^{-47}$ |

Table 7: ANOVA p-values for within-model variation across four different prompts in each setting. The bottom row shows ANOVA p-values for ToF scores across all models. p-values smaller than 0.005 is considered highly significant.

as examples when prompting for each target scenario. For each scenario, we include two examples—arranged in the order: Sycophantic, Non-Sycophantic, Non-Sycophantic, and then Sycophantic—to avoid introducing patterns that might bias the model’s response. The specific prompts are written in Figure 11-13.